

Decoupling Shape and Texture in SAM-2 via Controlled Texture Replacement

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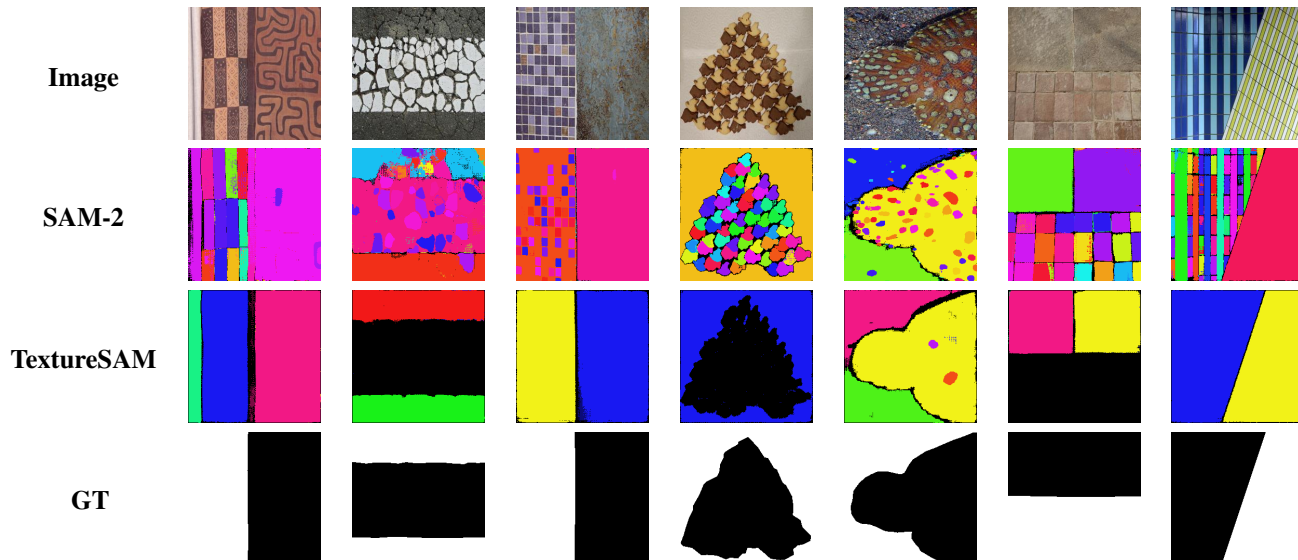


Figure 1. **From Shape Bias to Texture Awareness.** SAM-2 (second row) over-segments texture dominant scenes because it is shape biased. We propose a procedure to reduce the shape bias by fine-tuning it via controlled texture replacement using existing semantic masks. The fine-tuned model, that we term *TextureSAM*, produces coherent, texture-aligned regions (third row). This fine-tuning procedure corrects SAM-2’s failure modes and captures boundaries defined by texture transitions. These results are shown on the Real-World Textured Dataset (RWTD), and demonstrate significant improvements in segmenting natural textures without semantic cues.

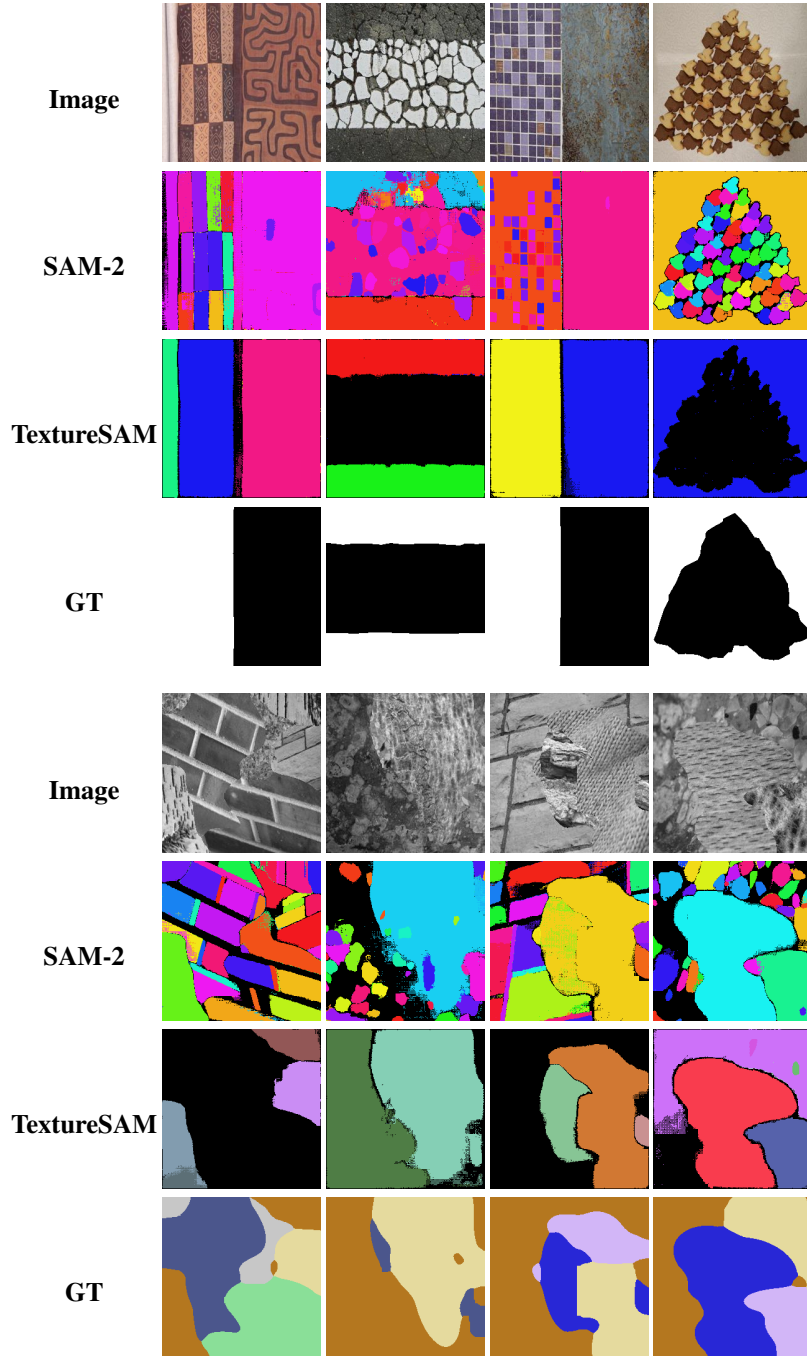


Figure 2. Qualitative results on STMD real (top) and synthetic (bottom) samples.

Abstract

Segment Anything Models (SAM) have demonstrated strong generalization in object segmentation across diverse datasets. However, their training on large-scale semantic segmentation data induces a shape bias, leading to

over-segmentation in texture-dominant scenes and severely limiting performance. This limitation is particularly pronounced in domains such as remote sensing and metallographic imaging, where meaningful boundaries are defined by texture variations rather than semantic structure. In this study, we investigate SAM’s shape bias and show that a simple fine-tuning strategy – based on incremental

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texture augmentations of semantically labeled data – can effectively calibrate this bias and guide the model toward texture-aware segmentation. By interpolating and replacing textures within **semantically** labeled regions, we generate texture-diverse instances of the same semantic category, enabling effective fine-tuning without requiring additional manual annotations. We release both the texture-oriented variant of SAM (“TextureSAM”) and the texture-augmented dataset used in our experiments to support reproducibility and facilitate further research on shape–texture bias in foundation models. We show that this fine-tuning approach mitigates SAM-2’s shape bias, improving segmentation performance on both real-world (RWTD, +0.20 mIoU) and synthetic (STMD, +0.18 mIoU) texture segmentation benchmarks. Our method also demonstrates improved zero-shot generalization on in-the-wild datasets, spanning aerial imagery, satellite scenes, and metallographic microstructures, nearly doubling SAM-2’s mIoU in domains where annotations align with texture boundaries. Our code and data are available [here](#).

1. Introduction

Conventional segmentation models are designed to recognize objects based on semantic features, yet many real-world applications rely on texture — a concept that is inherently difficult to define. Texture can manifest as structured, repeating patterns or as irregular, stochastic variations, making it challenging to model within a unified framework.

Powerful foundation models like SAM [21], and its successor SAM-2 [29], are biased toward shape-based representations rather than texture, and leveraging them for texture-sensitive tasks requires careful calibration of this inherent shape–texture bias. Traditionally, this typically requires fine-tuning on large, task-specific datasets with texture-oriented annotations. This challenge is particularly evident in materials science [2,6,31,32], where metallography and composite analysis requires texture-based segmentation to detect grain structures, fractures, and surface irregularities, as these features lack clear semantic categories. In remote sensing, texture is essential for land cover classification and vegetation mapping that can be represented as whole textured regions [4,15,23,25].

Figure 1 shows the problem and our proposed solution. The first row shows some sample images from the Real-World Textured Dataset (RWTD). Row two shows the segmentation results of SAM-2. As can be seen, SAM-2 exhibits over-segmentation due to its shape-dominant inductive bias (e.g., separating individual bricks within a wall, instead of recognizing it as a unified textured shape). Our solution, termed *TextureSAM*, is based on a new fine-tuning protocol that is able to calibrate the shape-texture bias of SAM-2. This leads the model to achieve much better tex-

ture segmentation results, as can be seen in row three. Row four shows the Ground Truth labeling, for reference.

Why does SAM-2 fail? We hypothesize that SAM-2 implicitly couples shape and texture cues — that is, it learns associations between object categories and their canonical textures. We quantify this behavior on dedicated texture datasets, and propose a method to decouple shape and texture through controlled fine-tuning, using existing annotations and texture-replaced images. This enables systematic calibration of SAM-2’s bias and allows us to study how texture information can be explicitly incorporated into segmentation. This highlights a fundamental research gap: *can a generic foundation model for segmentation be modified so that it explicitly incorporates texture information rather than relying solely on shape-based priors?*

Foundation models like SAM, while powerful, exhibit this pronounced shape bias, and no methodology currently exists to systematically expose and calibrate this bias. We address this by introducing a calibration protocol that gradually shifts the model’s focus from shape to texture, without requiring any new annotations. Our key insight is to leverage existing semantic segmentation datasets, but progressively replace object textures while preserving shape labels. This allows us to probe the shape–texture trade-off in a controlled setting and to modulate SAM’s inductive bias through targeted fine-tuning. The result is a principled framework, including a ready-for-use fine-tuning dataset, for analyzing and calibrating texture sensitivity in segmentation foundation models.

To achieve gradual texture replacement, we rely on a recently introduced compositional neural texture approach (CNT) [34] that can interpolate between source and style (i.e., texture) images. This way, we can rely on ground truth masks of existing datasets (ADE20k in our case) to dress instances with arbitrary new texture (from Describable Textures Dataset, or DTD [5], in our case), thus decoupling shape from texture. We finetune SAM-2 using the texture-decoupling dataset. We refer to this calibrated model as *TextureSAM* and use it to evaluate the effectiveness of our bias modulation protocol. We detail our calibration process and evaluate the resulting model on a number of datasets, also demonstrating utility in improving zero-shot segmentation ability on texture-heavy real world datasets. To our knowledge, this is the first approach to explicitly probe and guide a foundation segmentation model’s reliance on texture cues.

Research Questions. This work investigates the following key research questions:

- **RQ1:** To what extent is SAM-2 biased toward shape over texture, and how can this bias be revealed and quantified?
- **RQ2:** Can this bias be calibrated, without the need for a large-scale manually annotated texture oriented-dataset?

Contributions. The key contributions of this work are as follows:

- We present a general framework for calibrating the shape-texture bias in segmentation foundation models via texture-guided fine-tuning, **requiring no additional manual annotations**.
- We introduce a texture-augmented version of ADE20K (Textured-ADE20K), generated through controlled texture replacement within semantic regions, which enables disentangling shape and texture during training.
- We empirically demonstrate that SAM-2 can be calibrated toward texture-aware segmentation through this approach, yielding improved zero-shot performance in texture-centric domains.
- We release the texture-augmented dataset and the fine-tuned model checkpoint to support reproducibility and future work on shape–texture bias calibration.

Relation to prior work on shape vs. texture bias. Prior studies focus on classification, often increasing shape bias to improve robustness. In contrast, we address segmentation, and propose the first method to explicitly reduce shape bias in SAM-2 via a texture calibration protocol and dataset. Whereas most works are diagnostic in nature, our approach offers actionable fine-tuning strategies and reusable tools for future work.

2. Prior Work

2.1. Texture Analysis

Texture analysis is often associated with semantic segmentation and can be roughly divided into two parts: Semantic Segmentation and Instance Segmentation. In Semantic Segmentation, each pixel is assigned a label of the class (i.e., road, vehicle, sky, person, etc.) it belongs to. In Instance Segmentation, each pixel is assigned a label of the class and the instance it belongs to (i.e., two different people will be assigned different labels). Recently, the two have been combined into a single, holistic approach, termed Panoptic Segmentation, where the goal is to assign each pixel its class label, and in case there are multiple instances of the object, assign the different objects a different label. Panoptic Segmentation is used for various applications such as Autonomous Driving and general image understanding.

Another closely related branch of texture analysis is that of Boundary Detection, where the goal is to detect meaningful boundaries/interfaces between textures, cells, crystallites, etc. This branch evolved from the problem of edge detection, where edge detection is concerned with detecting changes in intensity values, while boundary detection is concerned with detecting boundaries between regions (i.e., textures).

Traditional methods mostly focus on low-level cues. For example, Martin *et al.* [24] proposed a probabilistic boundary (Pb) detection module that learns to detect boundaries in images using local brightness, color, and texture cues. Dollár and Zitnick [8] use Structured Forests to map local image patches to local edge maps.

A sequence of papers [3, 22, 33] suggested using deep features to improve image boundary detection. Common to all is the use of deep features as the space in which to detect boundaries. Exploiting multi-scale representations and dilated convolutions helped He *et al.* [13] push the state-of-the-art in the field even further. Recently, Pu *et al.* adapted Transformers for boundary detection [28]. Their approach is based on a two-level Transformer architecture, where the first level captures global scene information, and the second refinement level calculates the boundaries. Their network achieves state-of-the-art results on several datasets [19], including the BSD dataset [1]. The latest advancement using Transformers for this task is the Segment Anything Model (SAM) [21], which presents a significant step forward while also having limitations in professional data, especially in textural and medical imaging [14, 17].

2.2. SAM’s Limitations with Texture

Segment Anything (SAM) [21] introduced a new paradigm in segmentation research by enabling zero-shot segmentation across diverse datasets. SAM is trained on the SA-1B dataset, which contains one billion masks from eleven million images, using a Vision Transformer (ViT) backbone for feature extraction. SAM-2 [29] extends this approach with a larger dataset and architectural refinements but remains fundamentally designed for semantic segmentation rather than texture-aware segmentation.

The trade-off between shape and texture bias in deep learning has been extensively studied. Convolutional neural networks (CNNs) exhibit a strong texture bias due to their local receptive fields and weight-sharing properties [12]. In contrast, vision transformers (ViTs), such as those used in SAM, prioritize global shape over local texture [27]. This is due to their patch-based tokenization and self-attention mechanisms, which emphasize long-range dependencies at the cost of fine-grained texture representations [9]. Experiments using Stylized ImageNet [27] confirm that ViTs retain object recognition even under extreme texture alterations, further reinforcing their preference for shape-based segmentation. Additionally, real-world evaluations of SAM-2 [18] reveal that its performance degrades in texture-sensitive scenarios, underscoring the need for improved texture awareness in segmentation foundation models.

Recent work has explored methods for modulating shape versus texture bias in vision models. Gavrikov *et al.* [11] demonstrated that shape bias in vision-language models can be controlled via language prompts but, to our knowledge,

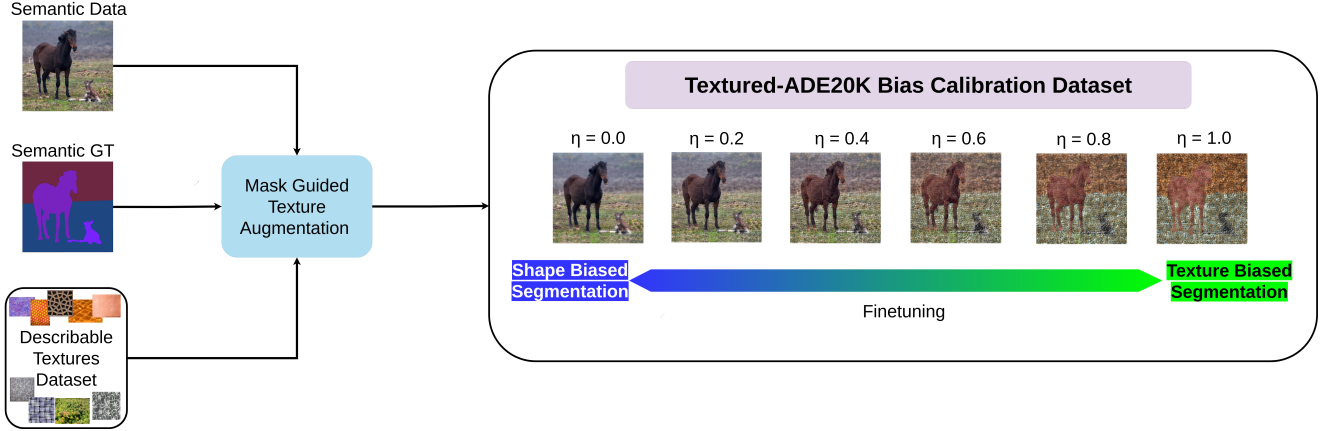


Figure 3. **Controlled Texture Augmentation Pipeline for Shape-Texture Bias Calibration.** Semantic data, ground truth masks, and a diverse texture dataset are combined via mask-guided texture augmentation to generate the Textured-ADE20K dataset. By progressively increasing the mixing parameter η , object textures are replaced in a controlled manner while preserving semantic structure. This enables fine-tuning a segmentation model from shape-biased ($\eta=0$) to texture-biased ($\eta=1$) behavior, facilitating systematic calibration of inductive bias in foundation segmentation models.

no prior work has explicitly addressed increasing texture bias in segmentation foundation models. While texture transfer techniques such as [34] enable controlled modification of textures, they have not been used to systematically fine-tune a foundation model for texture-aware segmentation. This work introduces a texture-focused adaptation of SAM, directly addressing its shape-prior limitation.

3. Controlled Texture Augmentation and Model Adaptation

3.1. Overview

To calibrate SAM-2’s shape–texture bias, we fine-tune it on a texture-augmented version of ADE20K using controlled, mask-aware texture replacement. Although SAM-2 excels at general-purpose segmentation, its strong reliance on high-level shape priors leads to failure in texture-dominant scenarios, often resulting in over-segmentation of perceptually uniform regions (separating individual texture elements). To address this limitation, we introduce a fine-tuning protocol that gradually replaces object textures within semantic masks using samples from the Describable Textures Dataset (DTD) [5], a collection of 5,640 real-world texture images. The degree of replacement is controlled by a mixing parameter $\eta \in [0, 1]$, allowing us to systematically vary the strength of texture cues during training. At low values of η , semantic structure is preserved, while high values fully overwrite object appearance, forcing the model to rely on texture rather than shape. The pipeline of our method is presented in Figure 3.

We train two variants: one with mild texture augmentation ($\eta \leq 0.3$, i.e. finetuning using four augmentation levels

0.0, 0.1, 0.2 and 0.3) and another with strong augmentation ($\eta \leq 1.0$, finetuning using all 11 augmentation levels), enabling us to study the shape–texture trade-off under different levels of texture emphasis. We evaluate both models on two texture-centric benchmarks: RWTD [20], a natural image dataset with annotated texture boundaries, and STMD [26], a synthetic dataset with clean texture transitions and no semantic structure. Segmentation performance is measured using mean Intersection over Union (mIoU) and Adjusted Rand Index (ARI), with particular attention to fragmentation and region coherence in texture-driven scenes.

3.2. Datasets

The ADE20K dataset [35] is a large-scale scene parsing dataset containing over 20,000 images with 150 semantic categories, including objects, parts, and materials. It provides densely annotated scene layouts, making it a benchmark for semantic segmentation, scene understanding, and contextual reasoning. The dataset covers a diverse range of indoor and outdoor environments, making it widely used in deep learning-based segmentation research. The training part of this dataset was used to train TextureSAM, and the validation part of the ADE20K dataset, comprising 2000 images, is used to assess our model’s ability to perform general semantic segmentation after fine-tuning.

Dataset Preparation To finetune SAM-2 while preserving as much of its original segmentation capabilities, we use the ADE20K training set which is a subset of SA-1B, the original data used in SAM-2’s training. This approach mitigates catastrophic forgetting [10], ensuring that Tex-

tureSAM retains general segmentation ability while adapting to texture-based cues. ADE20K presents a particularly challenging benchmark, with the original SAM-2 achieving only 0.46 mIoU on its validation set. The following section details the augmentation strategy applied to the ADE20K dataset in order to introduce texture awareness in SAM-2.

3.3. Textured-ADE20K dataset

We employ the texture transfer technique in [34] to create the Textured-ADE20K dataset. We transfer textures I_t from describable texture dataset (DTD) [5] to semantic content images I_c from ADE20K [35]. There are 5640 texture images organized into 47 categories of textures in the DTD dataset. ADE20K dataset contains 27574 images (25574 for training and 2000 for validation), where each image comes with associated ground-truth instance segmentation masks $\mathcal{M}$. To create the textured-ADE20K dataset for segmentation, the instances in an image are transferred with different textures. Figure 3 in the supplemental shows samples from the textured-ADE20K dataset.

Figure 2 in the supplemental gives an overview of the method. To texturize an image I_c , for each instance mask m , we randomly select a texture I_t from a unused category in the DTD dataset (if all categories are used, randomly select a texture from the entire dataset). Before further processing, we scale the content image I_c into $8l \times 8l$ pixels and the texture image I_t into $l \times l$ pixels. Since the model focuses solely on texture, we extract overlapping patches I_c^p of size $l \times l$ with spacing of $3/4l$ pixels from I_c , $p \in \mathcal{P} = \{0, 1, 2, \dots, \#patches\}$ – treating every patch as an independent texture sample. I_c^p is then encoded into a composition of Gaussians $\mathcal{G}_c^p$, and these representations are merged in their original spatial order to produce Gaussians $\mathcal{G}_c = \text{Merge}(\bigcup_{p \in \mathcal{P}} \mathcal{G}_c^p)$ reconstructed as the original image. Merge merges overlapping patches of Gaussians while ensuring smooth transition (see [34] for details). Each texture I_t is also encoded as Gaussians $\mathcal{G}_t = \{g_t\}$ by directly forwarding them through the encoder E . To alter texture within a binary instance mask $m \in \mathcal{M}$, we select Gaussians centered within the mask $\mathcal{G}_c^m = \{g_c^m\} = \{g_c | g_c \in \mathcal{G}_c, m(\mu_c) = 1\}$ and modulate their appearance features $f_c^m \in g_c^m$. For each f_c^m , we randomly pick a feature $f_t \in g_t$ from $\mathcal{G}_t$, the modified feature $\tilde{f}_c^m$ is computed as

$$\tilde{f}_c^m = \eta f_c^m + (1 - \eta) f_t. \quad (1)$$

The interpolation coefficient η determines the tradeoff between the input image and the texture image. When $\eta = 0$, $\tilde{I}_c$ is the *texturization* of I_c : $\tilde{I}_c$ closely matches I_c but is not the exact reconstruction (Figure 3 in the Supplemental, rows: 1-2, columns: 1-2). This is because we encode I_c as Gaussians using E patch-by-patch and reconstruct them using D , which (1) projects each image patch

onto the learned texture space (2) merges overlapping Gaussian patches and (3) reconstruct Gaussians as $\tilde{I}_c$. Intuitively, this process 1) finds closest texture embeddings for image patches 2) spatially merge texture embeddings to ensure smooth transition and 3) reconstruct images from blended textures.

3.4. Model Finetuning

The training protocol we use for finetuning SAM-2, follows the default finetuning configuration from the SAM-2 repository, ensuring consistency with prior work. Hyperparameters remain unchanged and are provided in the Supplemental Materials for reference. Due to resource constraints, we finetune the sam2_hiera_small variant of SAM-2 using a single A100 GPU. Despite the reduced model size, this configuration allows us to efficiently evaluate the impact of texture augmentation while maintaining alignment with the original training setup. We train for 19 epochs on $\eta \leq 0.3$ and 25 epochs on $\eta \leq 1.0$, balancing training time with performance improvements.

3.5. Evaluation Protocol

Metrics We measure segmentation performance using two complementary metrics:

- **Mean Intersection over Union (mIoU):** Evaluates the overlap between predicted and ground truth regions.
- **Adjusted Rand Index (ARI):** Measures clustering consistency, particularly useful for assessing segmentation quality in texture-based datasets, as it penalizes fragmentation of textured regions to individual perceptual elements.

We compare TextureSAM ($\eta \leq 0.3$ and $\eta \leq 1.0$) against the original SAM-2, analyzing how well the model adapts to texture cues while maintaining general segmentation capability.

Inference and Evaluation Procedure. To ensure meaningful segmentation results, we modify the default inference parameters of SAM-2 for TextureSAM inference. Specifically, we modify the `points_per_side` parameter from 32 to 64, and the `stability_score_thresh` from 0.95 to 0.2. Using the default inference parameters for TextureSAM resulted in no predicted masks for most images, making direct evaluation unreliable. By adjusting the working point of the model inference we allow for a more meaningful comparison. Given that the modified inference parameters lead to a more dense segmentation which may increase textured area fragmentation, we ensure a fair evaluation by obtaining results for the original SAM-2 model with both the original

and modified inference parameters. Throughout this paper, we refer to the original model with default parameters as **SAM-2**, and to the model evaluated with our adjusted inference parameters as **SAM-2***.

4. Results

We evaluate the impact of our texture bias calibration on segmentation performance using both texture-oriented and general-purpose benchmarks.

Our primary benchmarks are the Real-World Texture Dataset (RWTD) and the Synthetic Textured Masks Dataset (STMD). RWTD consists of 256 natural images with ground truth annotations marking texture boundaries, not semantic object edges [20]. In contrast, STMD contains purely synthetic images composed of texture transitions, deliberately excluding semantic content [26]. Together, these datasets allow us to test whether our fine-tuned models better capture texture-defined regions, even in the absence of clear object structures.

To assess the generalization of our method, we additionally evaluate on three non-texture-specific datasets, which include naturally occurring texture cues but are not explicitly annotated for texture. These datasets are described in detail in the supplemental material. We compare TextureSAM (with calibration levels $\eta \leq 0.3$ and $\eta \leq 1.0$) to the unmodified SAM-2, using both its default inference settings and those adapted for TextureSAM to ensure a fair comparison.

Finally, to assess whether texture calibration compromises semantic segmentation performance, we evaluate on the original ADE20K validation set. The ADE20K validation set is used to evaluate the overall segmentation quality of the bias calibrated TextureSAM variants and vs. the original SAM2 model. For this evaluation we apply mask aggregation, where predicted segmentation masks are grouped based on their overlap with ground truth regions. This approach provides a more holistic measure of segmentation performance by consolidating fragmented predictions that belong to the same texture-defined region. As a complementary analysis, we also report metrics without aggregation, where individual predicted masks are evaluated independently. This penalizes models that over-segment regions by producing multiple small fragments instead of a single coherent mask for a textured region.

Below, we detail our findings across these datasets and evaluation protocols.

4.1. Synthetic STMD Dataset Results

The synthetic texture-shift dataset provides a controlled benchmark where segmentation is based purely on texture, with no object structures present. This allows us to isolate the effect of texture on segmentation performance without interference from shape-based cues. As shown in Table 1,

TextureSAM with $\eta \leq 1.0$ achieves the best mIoU and ARI scores, demonstrating improved segmentation performance on pure textures. The $\eta \leq 1.0$ training includes images that are defined solely by texture borders, similar to the images in the STMD, making this result logical. The $\eta \leq 0.3$ performance falls between the original model and the $\eta \leq 1.0$, validating the proposed method’s ability to calibrate the reliance of the fine-tuned model on texture signals via the mixing parameter.

The original SAM-2 exhibits significantly lower ARI and mIoU scores, confirming that it tends to over-segment textured regions into multiple smaller components due to its inherent preference for shape-based segmentation. In contrast, the calibration procedure mitigates this fragmentation, leading to more consistent segmentation by TextureSAM variants that adhere to texture boundaries rather than arbitrary shape structures. Further visualizations, including a box-plot of the number of masks predicted by SAM-2 are presented in the supplemental material, depicting a much lower number of masks predicted for the fine-tuned model in comparison with the original SAM-2.

4.2. Real-World Textured Dataset Results

On the Real-World Textured Dataset (RWTD), which comprises natural images, we observe a consistent improvement in segmentation performance with TextureSAM compared to SAM-2, affirming that the calibration procedure was effective at shifting the model’s focus toward texture. As RWTD is specifically designed for texture-based segmentation, this dataset provides a strong benchmark for evaluating how well models can capture natural texture-defined regions rather than relying on shape cues. Figure 1 provides a qualitative visualization of the difference between the original SAM-2 model and TextureSAM. It can be seen that SAM-2 tends to fragment textured regions based on semantic elements comprising the texture, while TextureSAM produces more coherent segmentations aligned with texture boundaries. The supplemental material includes further visualizations, predictions on the entire RWTD dataset, as well as a box-plot of the number of masks predicted by SAM-2 to demonstrate the over-segmentation failure mode of the SAM-2 model on real textured data. Quantitative results are presented in Table 1, where TextureSAM with $\eta \leq 0.3$ achieves the best mIoU and ARI scores. Different to STMD, this dataset is not comprised of pure textures, causing the $\eta \leq 1.0$ to perform worse than the milder $\eta \leq 0.3$ variant, further demonstrating the efficacy of the calibration procedure.

4.3. ADE20K Validation Dataset Segmentation Results

Finetuning SAM-2 to be more texture-aware comes at the risk of damaging its existing semantic segmentation

Model	STMD		RWTD	
	mIoU	ARI	mIoU	ARI
SAM-2	0.07	0.16	0.26	0.36
SAM-2*	0.17	0.16	0.14	0.19
TextureSAM				
$\eta \leq 0.3$	0.33	0.32	0.47	0.62
$\eta \leq 1.0$	0.35	0.34	0.42	0.54

Table 1. Segmentation results on the STMD synthetic dataset and the RWTD natural image dataset. SAM-2* indicates SAM-2 evaluated using the inference parameters of TextureSAM. TextureSAM trained with texture augmentations significantly outperforms the original SAM-2 across both datasets. The model with mild augmentations ($\eta \leq 0.3$) performs best on RWTD, while the version with strong augmentations ($\eta \leq 1.0$) achieves the highest scores on STMD.

capabilities. We use ADE20K in an attempt to avoid catastrophic-forgetting, as it is a part of the original dataset used to train SAM-2. ADE20K’s validation dataset is a challenging semantic segmentation benchmark, even for the SAM-2 original model. The validation dataset is therefore useful to quantify the cost of the texture bias calibration procedure to the semantic segmentation capabilities of the original SAM-2 model. For each ground truth (GT) mask, we first identify overlapping predicted masks in the model’s output and unify them before calculating IoU. This provides an evaluation of the model’s overall segmentation ability, disregarding fragmentation. We report results both with mask aggregation and without. Table 2 shows a comparison of the original SAM-2, with both default and TextureSAMs inference parameters, as well as TextureSAM mild augmentation model ($\eta \leq 0.3$) and severe augmentation ($\eta \leq 1.0$). It can be observed that both variants of TextureSAM achieve comparable, and slightly higher scores in ARI and mIoU compared to SAM-2. The Original SAM-2 with the modified inference parameters scores the highest in the aggregated mIoU score, which is the most relevant for semantic segmentation evaluation. TextureSAM with mild augmentations $\eta \leq 0.3$ achieves a lower (-0.1 mIoU) score indicating that shifting the model’s focus towards texture did reduce semantic segmentation capabilities. As can be expected, we see that using a higher *eta* value hinders the semantic segmentation further (-0.15 mIoU). Using the modified inference parameters increased the original SAM-2’s score by producing more masks. This increase was also recorded for the texture datasets. Qualitative visualization for the ADE20K validation dataset are included in the supplemental.

ADE20K Results	mIoU	ARI	mIoU,Aggr.
SAM-2	0.08	0.08	0.46
SAM-2*	0.1	0.11	0.65
TextureSAM $\eta \leq 0.3$	0.11	0.11	0.55
TextureSAM $\eta \leq 1.0$	0.12	0.11	0.40

Table 2. Results for the ADE20k natural image dataset. SAM-2* indicates using SAM-2 with the parameters used for TextureSAM. SAM-2* scores the highest in the aggregated mIoU score (mIoU,Aggr.), which is the most relevant for semantic segmentation evaluation. TextureSAM with mild augmentations $\eta \leq 0.3$ achieves a lower (-0.1 mIoU) score indicating that shifting the model’s focus towards texture did reduce semantic segmentation capabilities. Using the modified inference parameters increased the original SAM-2’s score by producing more masks. For the default inference parameters SAM-2 the performance is comparable to the $\eta \leq 0.3$ model

Model	POTSDAM		UCHSDB		Satellite	
	mIoU	ARI	mIoU	ARI	mIoU	ARI
SAM-2	0.14	0.15	0.08	0.07	0.08	0.11
SAM-2*	0.13	0.14	0.07	0.09	0.07	0.08
TextureSAM						
$\eta \leq 0.3$	0.21	0.20	0.25	0.23	0.17	0.22
$\eta \leq 1.0$	0.22	0.21	0.26	0.26	0.13	0.12

Table 3. Performance comparison of SAM-2, SAM-2*, and TextureSAM across three texture-centric datasets: POTSDAM (aerial urban scenes), UCHSDB (ultrahigh carbon steel microstructures), and Satellite imagery. Metrics reported are mean Intersection over Union (mIoU) and Adjusted Rand Index (ARI). TextureSAM demonstrates consistent improvements in both metrics, with $\eta \leq 1.0$ yielding the strongest overall performance. SAM-2* refers to SAM-2 evaluated using TextureSAM’s inference configuration. This demonstrates the viability of the texture bias calibration process in real world scenarios by reduction of the over-segmentation failure mode.

4.4. Texture Segmentation in the Wild

Our work focuses mainly on the viability of the shape-texture calibration of SAM-2. Here we would like to briefly demonstrate the utility and generalization potential of this process to spur the community interest in this line of future research. We conduct zero-shot inference experiments on three datasets which may benefit from a texture-aware variant of SAM-2: ISPRS Potsdam [30] (urban aerial imagery), UHCSDB (UltraHigh Carbon Steel micrograph DataBase) [7], and the Dubai Aerial Imagery Dataset [16], that includes satellite imagery. These datasets differ significantly from the training data, and are not texture-specific as the dataset used to probe the calibration procedure’s effectiveness. **Potsdam** includes high-resolution, multi-class

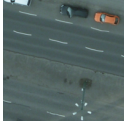
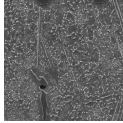
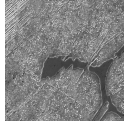
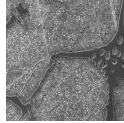
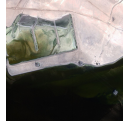
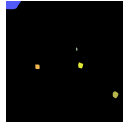
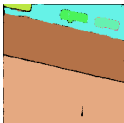
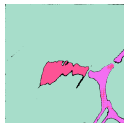
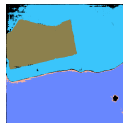
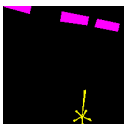
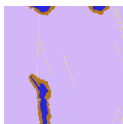
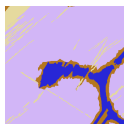
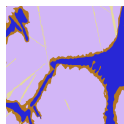
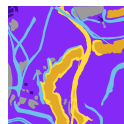
	POTSDAM			UHCSDB			Satellite		
	Ex. 1	Ex. 2	Ex. 3	Ex. 1	Ex. 2	Ex. 3	Ex. 1	Ex. 2	Ex. 3
Image									
SAM-2									
	mIoU: 0.001	mIoU: 0.110	mIoU: 0.231	mIoU: 0.108	mIoU: 0.054	mIoU: 0.013	mIoU: 0.046	mIoU: 0.049	mIoU: 0.116
SAM-2*									
	mIoU: 0.119	mIoU: 0.086	mIoU: 0.159	mIoU: 0.065	mIoU: 0.075	mIoU: 0.043	mIoU: 0.097	mIoU: 0.113	mIoU: 0.070
TextureSAM									
	mIoU: 0.223	mIoU: 0.139	mIoU: 0.275	mIoU: 0.164	mIoU: 0.314	mIoU: 0.247	mIoU: 0.149	mIoU: 0.228	mIoU: 0.123
GT									

Figure 4. **Qualitative comparison across three datasets** (POTSDAM, UHCSDB, Satellite). Each dataset shows three representative examples (Ex. 1–3). TextureSAM consistently delineates texture-defined regions more faithfully than SAM-2 and SAM-2*. mIoU values are displayed beneath each prediction. In all cases, we nearly double the mIoU score.

satellite scenes with structured man-made urban patterns, which challenge shape-aware models to capture texture-defined boundaries. Similarly, the satellite imagery dataset includes both natural and man-made patterns which may benefit from texture aware segmentation. **UHCSDB** contains microscopy images of complex metal microstructures with irregular texture transitions and no clear semantic objects. Table 3 provides quantitative results, and Figure 4 offers a qualitative visualization of the segmentation. Evaluating the fine-tuned model on these three texture-centric datasets: metallography (UHCSDB), aerial drone imagery (Potsdam), and satellite imagery, we observe improved zero-shot segmentation performance over SAM-2, with gains of +0.17, +0.08, and +0.09 mIoU, almost doubling the original SAM-2 scores. On ISPRS Potsdam (aerial urban scenes), TextureSAM better captures man-made textures like rooftops and roads, improving mIoU by +0.08 over SAM-2. In UHCSDB (metallographic micrographs), it segments complex grain structures more coherently, avoiding over-fragmentation and achieving a +0.17 mIoU gain. On Dubai Satellite Imagery, TextureSAM delineates surfaces such as vegetation and unpaved land more faithfully, with a +0.09 mIoU increase. This validates the generaliza-

tion and practical relevance of our approach, indicating that the bias calibration method presented here may be useful for leveraging foundation models for segmentation for real-world texture-driven applications.

5. Conclusions

With regard to RQ1, we have demonstrated and quantified SAM-2’s shape-bias, via evaluation on texture-centric datasets. We provide strong empirical evidence that agree with previous works, supporting the hypothesis that SAM-2 is shape biased. This is evident as failure mode characterized by over-segmentation of textured regions. To address RQ2, we introduce a bias calibration protocol, leveraging existing semantically annotated segmentation data, and applying controlled texture replacement, to produce a “texture-decoupling” fine-tuning dataset. We show that finetuning SAM-2 on this dataset successfully mitigates its shape bias, leading to improved segmentation in texture-driven scenarios. We provide empirical evidence that it is indeed possible to shift a foundation model towards performing texture-driven segmentation. These observations highlight a trade-off between enhancing texture sensitivity and preserving broader segmentation performance, high-

lighting the merit of continued research into controllable calibration of texture-shape bias.

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